

Deloitte MSME Valuation

Touchless Comparable-Company Discovery & Valuation Platform

Final Production Architecture — Realistic, Implementable, Defensible

Classification	Confidential — Internal Architecture Document
System type	Fully automated (touchless) — no human in the loop
Primary use case	Indian private MSME / SME valuation via listed comparable companies
Discovery source	Dun & Bradstreet (paid) — company universe + economic profiles
Financial enrichment	Screener (free), Tofler (free), yfinance (free backup)
Business understanding	D&B descriptions + company website (Selenium) + LLM extraction
Valuation method	Comparable-company multiples (EV/EBITDA, EV/Revenue) with disclosed discount
Output	Indicative valuation range (low / mid / high) with confidence score and evidence
Date	July 08, 2026

READ FIRST — WHAT “ACCURATE” MEANS HERE. For a private company with no public share price, no valuation system — human or automated — produces a single exact number. A defensible output is an indicative **range** derived from real listed-peer multiples, with a **confidence score** and a full evidence trail. This architecture is engineered to produce exactly that. Any system claiming a single precise private-company valuation should be treated with suspicion by a reviewer.

1

Data Sources — Role and Honest Reality Check

What each source can and cannot do

The architecture uses each source only for what it can realistically deliver. Discovery, financial enrichment, and business understanding are deliberately separated so that a weakness in one source does not corrupt the whole pipeline.

Source	Access	Role in System	What It Realistically Delivers	Known Limitation
Dun & Bradstreet	Paid	Primary discovery universe + economic profile	Company identity, industry codes, business description, size band, employee band, export indicator, corporate family. Search by industry + geography + size.	No free-text economic search. Description quality varies. Indian MSME coverage weaker than large caps — must verify per sector.
Screener	Free	Listed-peer financials + multiples	Historical financials, margins, ROE/ROCE, market cap, P/E, EV/EBITDA for listed Indian companies.	Listed companies only. Free tier: no bulk API — scraped per company. Rate limits apply.
Tofler	Free	Private company MCA enrichment	CIN, MCA filings, statutory financials where filed, directors, charges for the target and private peers.	Free tier is shallow — often only latest year, sometimes stale. Not a discovery source.
yfinance	Free	Backup listed-peer multiples	EV/EBITDA, EV/Revenue, market cap, sector for NSE/BSE listed peers when Screener is thin.	Unofficial library. Can break without notice. Listed only.
Company Website	Free (scrape)	Business understanding for economic profile	Products, customers, export markets, B2B/B2C signals — often richer than D&B for MSMEs.	Only if URL exists. Selenium needed for JS-heavy sites. Unstructured text.

DECISION: D&B is the discovery universe because it is the only paid source you have that can return a filtered candidate list. Screener and Tofler are enrichment only — they are queried **after** peers are selected, never for discovery. The company website is promoted to a first-class input because for Indian MSMEs it is frequently the single richest description of what the business actually does.

2

End-to-End Architecture

Two pipelines: offline universe build + online valuation

The single most important design decision: comparable discovery cannot happen live at query time. D&B cannot be asked “find companies whose operating model is B2B gold-jewellery manufacturing.” That economic search must run against a **pre-built index** of company economic profiles. The system therefore has two pipelines — an **offline pipeline** that builds and refreshes the searchable universe, and an **online pipeline** that runs at valuation time.

2.1 System Diagram

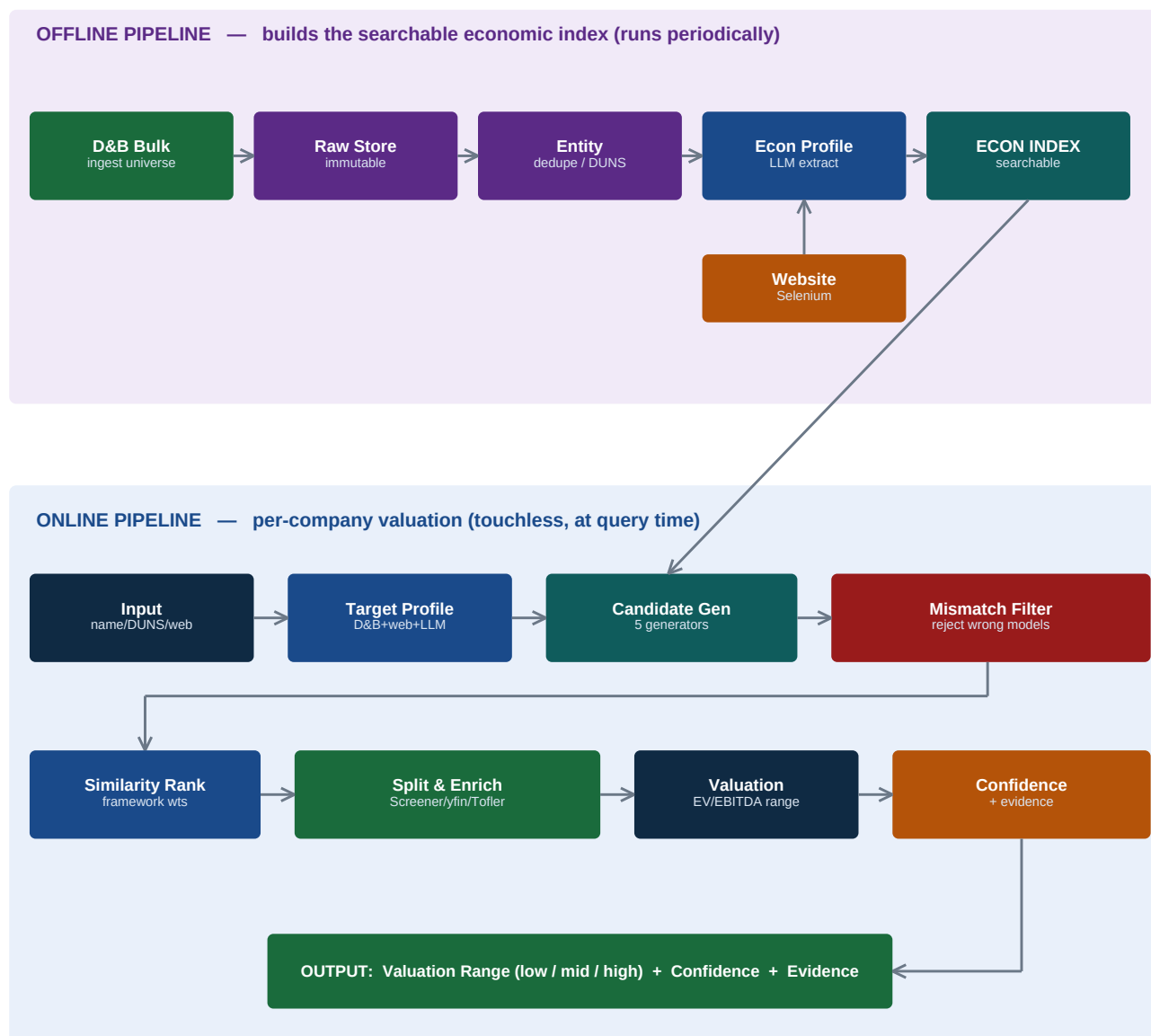


Figure 1 — Offline universe pipeline (top) builds the searchable economic-profile index. Online valuation pipeline (bottom) runs per company at query time.

3

Offline Pipeline — Building the Searchable Universe

Runs periodically, not at query time

This pipeline ingests the D&B company universe for the sectors you care about, builds an economic profile for every company, and stores it in a searchable index. It runs on a schedule (for example monthly) and is the foundation that makes fast, accurate discovery possible at query time.

Step	What Happens	Source / Tech	Output
O-1 Bulk ingest	Pull D&B companies filtered by target sectors, geography (India), and size bands.	D&B API (paginated)	Raw company records
O-2 Raw store	Store original D&B payload unchanged, with retrieval date. Never overwrite.	Object store / DB	Immutable raw evidence
O-3 Entity resolution	Deduplicate using DUNS as primary key; CIN + website + name+city as secondary.	Deterministic rules	Unique company entities
O-4 Economic profile	LLM extracts operating model, value-chain position, customer type, export flag, brand-owner flag from D&B description (+ website if available).	LLM + website scrape	Structured economic profile
O-5 Internal taxonomy	Map broad D&B class to fine internal node (e.g. Consumer Goods → Jewellery → Gold Jewellery Mfg).	Rule-based mapping	Taxonomy node per company
O-6 Index build	Store profile as structured fields + a text embedding for similarity search.	Vector + structured DB	Searchable economic index

REALITY CHECK: The offline pipeline is where the real engineering cost sits, and it is unavoidable. Without a pre-computed economic index, the five candidate generators have nothing to search and discovery degrades to raw industry-code matching — the exact failure the project set out to avoid. Budget for this as the core of the build, not an afterthought.

4

Online Pipeline — Per-Company Valuation

Runs at query time, touchless

When a company is submitted for valuation, this pipeline builds its economic profile, searches the pre-built index for comparables, filters and ranks them, pulls live financials for the top peers, and computes a valuation range — with no human intervention.

Step	What Happens	Source / Tech	Output
Q-1 Input	Accept registered company (name / DUNS / CIN / website) or MSME minimal manual input.	API	Target identity
Q-2 Target profile	Fetch D&B record; scrape website if URL; build target economic profile (same extractor as O-4).	D&B + Selenium + LLM	Target economic profile
Q-3 Candidate generation	Query the index via 5 generators: industry, product/service, operating model, scale, geography.	Economic index search	Candidate pool
Q-4 Mismatch filter	Hard-reject economic mismatches (e.g. retailer vs manufacturer, miner vs fabricator).	Rule filters	Clean candidate set
Q-5 Similarity ranking	Score with framework-specific weights (manufacturing / services / tech / financial).	Deterministic scoring	Ranked peers + component scores
Q-6 Split & enrich	Listed peers → Screener/yfinance for multiples. Private peers → Tofler for MCA financials.	Screener / yfinance / Tofler	Peers with financials
Q-7 Valuation engine	Median peer EV/EBITDA (or EV/Revenue fallback) × target metric; debt bridge; private discount; P25/P75 range.	Deterministic Python	Valuation range
Q-8 Confidence + report	Compute confidence from data completeness & peer quality; LLM narrates; output range + evidence.	Python + LLM	Final valuation report

5.1 Economic Profile — the structured object built for every company

This is what makes the gold-jewellery example work. Two companies in the same D&B code are separated by their economic profile — a manufacturer-exporter is matched to other manufacturer-exporters, not to retail chains or miners.

Field	Example value	Why it matters for peer selection
operating_model	manufacturer	Separates makers from traders, retailers, platforms
value_chain_position	manufacturing	Miner vs fabricator vs retailer sit at different points
customer_type	B2B / OEM	B2B and B2C carry different margins and multiples
export_driven	true	Exporters face different demand and FX profiles
brand_owner	false	Brand owners command premium multiples vs contract makers
scale_band	■100–500 Cr	Peers must be of comparable size to be valid
taxonomy_node	Gold Jewellery Mfg	Fine internal class, not broad D&B code

5.2 Framework-specific similarity weights

A single universal ranking formula fails. Each business type weights comparability differently:

Dimension	Manufacturing	Services	Technology	Financial
Operating model	30%	25%	20%	20%
Product / segment	25%	20%	30%	25%
Scale	20%	20%	15%	25%
Customer type	15%	25%	25%	10%
Geography	10%	10%	10%	20%

SCOPE DISCIPLINE: Build only these four frameworks for V1. Do not attempt twenty industry-specific formulas before the system has produced a single valuation. Four cover the vast majority of Indian MSMEs; add more only when real cases demand them.

6.1 Computation — pure Python, no LLM touches a number

Step	Formula / Rule
Primary multiple	Median EV/EBITDA of top listed peers × target normalised EBITDA
Fallback multiple	Median EV/Revenue × target revenue (used when EBITDA multiples unusable)
Equity bridge	Equity Value = Enterprise Value – Total Debt + Cash
EBITDA normalisation	Add back above-market owner salary & personal expenses (MSME-critical)
Private discount	20–30% illiquidity discount by size band, disclosed explicitly
Range	P25 / median / P75 peer multiples → low / mid / high valuation

6.2 Confidence score — the honesty mechanism

Every valuation carries a confidence score so a touchless output is never blindly trusted. It is computed deterministically from the factors below and shown prominently in the report.

Confidence factor	High confidence	Low confidence
Target data source	D&B + website + MCA all present	Only minimal manual input
Number of clean peers	5+ well-matched listed peers	Fewer than 3 peers
Peer financial freshness	Live listed multiples	Stale MCA filings only
Economic profile clarity	All fields extracted confidently	Sparse description, guessed fields
Multiple dispersion	Peer multiples tightly clustered	Wide scatter in peer multiples

REALITY CHECK: When confidence is low, the system must widen the range and say so — not hide it. A touchless system's credibility comes from disclosing its own uncertainty, not from producing a falsely precise single figure.

Reality Checks — Where This Fails and the Mitigation

Honest failure-mode analysis

Risk / Failure Mode	Likelihood	Mitigation Built Into Architecture
D&B thin for a given MSME sector	High	Website scrape + MCA fallback for target profile; confidence score drops and range widens automatically.
No listed peers exist for a niche business	Medium	Fall back to broader taxonomy node; use EV/Revenue; flag low confidence; widen range.
Entity resolution fails on common names	Medium	DUNS as hard primary key; require CIN or website match before merging; unmatched stays separate.
Screener free-tier rate limits / blocks	Medium	yfinance as backup multiple source; cache peer financials in offline refresh.
Stale private-peer MCA financials	High	Prefer listed peers for multiples; use private peers for context only; disclose vintage in evidence.
LLM mis-extracts economic profile	Medium	Store raw evidence (O-2) so profiles can be re-derived; confidence score reflects extraction certainty.
Touchless system over-trusted by user	High	Mandatory confidence score + explicit 'indicative range, not a formal valuation' disclaimer on every output.

7.1 What was deliberately cut to stay implementable

- **Human-in-the-loop review** — removed by requirement. Replaced by the confidence score and evidence trail as the trust mechanism.
- **Twenty industry frameworks** — cut to four for V1. Add on real demand.
- **Full entity-resolution ML** — deferred. DUNS-primary deterministic rules for V1.
- **Real-time discovery** — impossible against D&B; replaced by the offline economic index.

BOTTOM LINE: With paid D&B as the discovery universe and free Screener / Tofler / yfinance for enrichment, this system is realistically buildable and produces a **defensible indicative valuation range with a confidence score and full evidence trail** — touchless. It will not produce a single exact number for a private company, because no honest system can. The offline economic-profile index is the critical build investment; everything else is standard, well-understood engineering.